Mutriku wave power

Unpacking the full life-cycle footprint of breakwater wave energy

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Mutriku integrates oscillating water columns into a harbour breakwater in northern Spain. It combines electricity generation with turbine testing [1,4]. This case analyses Mutriku's environmental impacts across its full life cycle, asks how adverse effects can be reduced and whether improving output can justify the burdens of building, operating and retiring the plant.

How oscillating water column technology evolved

Sources: [7,8]

1910 — Bochaux Praceique builds the believed first oscillating water column (OWC)

1940s — Masuda develops a wave-powered navigation buoy

Late 1970s — Wells turbine invented: airflow both ways, no rectifying valves needed

1985 — Toftestallen, Norway: shoreline OWC prototype

1999–2000 — Pico (Azores) and LIMPET (Scotland) plants; LIMPET ran over 10 years

2011 — **Grid connection: 16 chambers × 18.5 kW (296 kW) [1,14]**

Mutriku integrates the OWC into a breakwater, so one structure also protects the harbour.

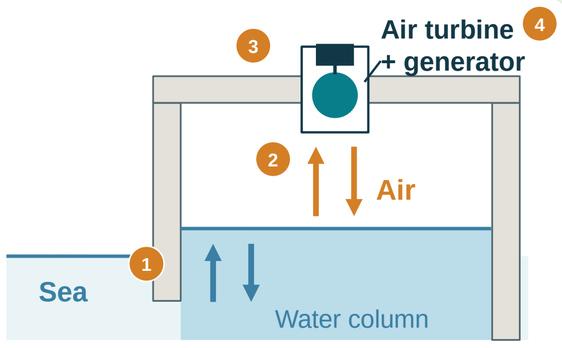
Did you know?

Europe's first: EVE describes Mutriku as the first commercial plant in Europe to generate electricity from waves [13].

Longest-running: EVE reports more operating hours since 2011 than any other wave plant of its kind [13].

Open to visitors: the plant is part of the tourist offer of the UNESCO-recognised Basque Coast Geopark [14].

01 How it works



Submerged opening

1. Wave lifts the water column
2. Air is pushed through the turbine
3. Wells turbine keeps turning one way
4. Generator makes electricity

16 × 18.5 kW = 296 kW [1,14]

Simplified section, not to scale. Arrows show alternating motion. Diagram based on [1].

02 Where it sits: location and settings



Photo: Txo (2026), Wikimedia Commons, CC0 1.0 [19]

Location: inside the harbour breakwater at Mutriku, on the Basque coast of northern Spain.

Why there: built into the breakwater, one structure shares harbour protection, port use and electricity generation [1,11]. The BiMEP Mutriku area also hosts turbine testing [4,9].

03 Where the carbon burden sits

226 g CO_2e/kWh — Life-cycle total, 20-year model [1]

? Dismantling emissions not quantified (set to 0)



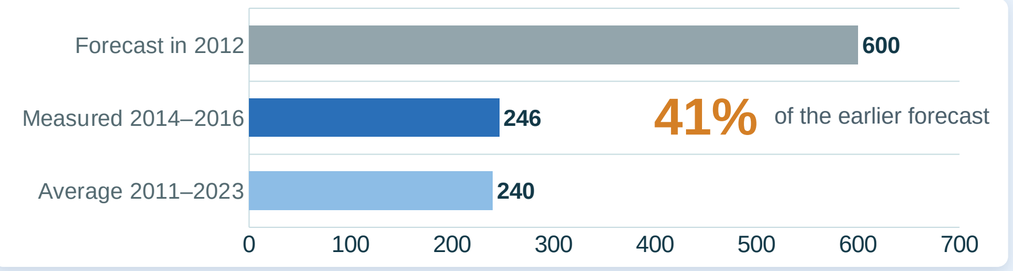
- **Construction 86.6%**
- **Operation & maintenance 5.5%**
- **End of life 7.9%**

Secondary inventories
Staff data
Dismantling set to 0 (no data)

Rounded stages sum to 225.3 vs 226 reported [1]. Model: 20 years, 4.93 GWh [1].

Construction dominates, so material demand is the priority. The model includes breakwater construction but excludes its maintenance [1]. Mutriku emits 53–94% more greenhouse gas than the other ocean devices compared in [1].

04 Output: forecast and evidence



41% of the earlier forecast

MWh/year [2,3]. Average: over 3 GWh [13,14] ÷ 12.5 years (author's calc.).

About 10 of the 14 turbines analysed working on average [2]

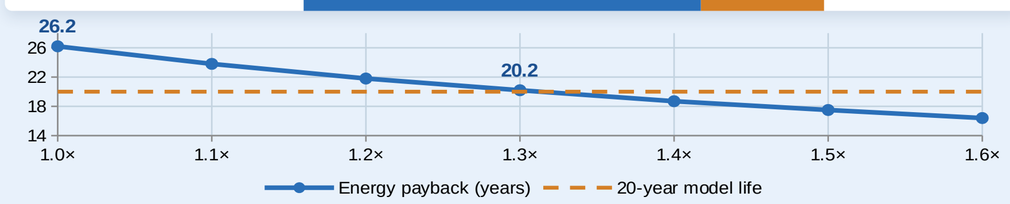
Cumulative generation reached 1 GWh in 2016 and 2 GWh in 2020 [17], then over 3 GWh by the end of 2023 [13,14], so the gap persists.

Beyond carbon: the wider footprint

▲ Favourable ▼ Adverse – Neutral

- ▼ Construction noise was heard 3–10 km away and worried locals [10]; one month of 2022 operating noise monitoring (one of three hydrophones failed) found no plant contribution to the soundscape [18]
- – Breakwater may alter sediment and flow; the OWC adds no significant change [10]
- ▲ Turbines sit in the air, not the sea [11]; plant sits on the breakwater, not the seabed [12]
- ▼ Salt exposure and splashing still cause corrosion and equipment failures [15]
- ▼ Most other LCA categories (e.g. marine ecotoxicity) are also driven by construction [1]
- ▼ Construction used about 12.9 kt of concrete and 21.3 kt of limestone [1]

05 Energy payback



Extend service life | **Raise net output** | **Use less material**

Break-even: about 31% more output (26.2 ÷ 20 = 1.31) or about 26 years' life. [1] quotes 8 years for solar PV and 1.5 for wind. Axis: output relative to model; author's scaling, embodied energy fixed [1].

06 Impacts, trade-offs and mitigation

▲ Favourable ▼ Adverse ? Not assessed

	Environment	Society
Build	▼ 195 g CO_2e/kWh [1]; hard-bottom habitat site, effects not assessed [1,10] ●	▼ Construction noise concerned local residents [10] ●
Operate	▲ Low emissions: 12.4 g CO_2e/kWh [1] ● ▲ No significant acoustic impact in 2016 [10]; no formal complaints in 13 years [14] ● ▼ Near-field sound and species effects unresolved; no rise seen ~1 km away [5,10] ●	▲ Electricity [1,14], turbine-testing access [4,9], harbour protection [11] ● ▲ Visitor interest suggests a local benefit [3,14] ● ▼ Public cost: €6.7M project, €2.3M for the energy plant [16] ●
Retire	? Habitat effects of removal not assessed; dismantling emissions modelled as zero [1] ● ▲ Steel recycling cuts metal depletion; more recycling is not always better overall [1] ●	? Social effects of closure (e.g. jobs) not assessed; acceptance data date from 2013 [6]] ●
Fix	Cut virgin materials · Verify marine durability · Upgrade turbines [4] · Compare end-of-life options	Coordinate work schedules · Track output and complaints · Residents shape decisions [5,6]

● Modelled ● Source-reported ● Unsourced or unresolved

Community milestones

Selected milestones, not a time-scaled axis.

- Jul 2011 – Plant opens [3]
- Dec 2011 – Storm shutdown [3]
- Mar 2012 – Operation resumes [3]
- 2013 – Social study [6]
- 2025 – SafeWAVE paper [5]

Selection prioritised recent (2018+) primary studies on impacts, output and turbines; pre-2018 sources [3,6,11,16] are used only where no newer Mutriku-specific study was found. This focused review is not systematic.

Conclusions and Recommendations

Key findings

1. **Construction dominates:** 86.6% of the 226 g CO_2e/kWh total, and the largest share in all nine LCA categories [1].
2. **Payback exceeds service life:** 26.2 years against a 20-year model life; output was ~41% of forecast [1–3].
3. **Evidence is incomplete:** dismantling emissions assumed to be zero for lack of data [1]; social evidence is dated [6].

Key judgement

Wave electricity alone does not make Mutriku sustainable: output would need to rise ~31%, or life extend to ~26 years, just to repay its embodied energy. Only lower material use, higher output or longer life, with full life-cycle accounting and fair sharing of burdens, can justify it.

1. Upgrade within the existing structure
2. Compare equivalent alternatives
3. Test lifetime, downtime, recycling
4. Approve only with full accounting
5. Publish monitoring; revisit

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